

iCOCKPIT - Jetro Mobile Client

Software Requirement Specifications

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1 Introduction

This section gives a general overview of everything included in this SRS and also provides you with definitions of specific terms used in this SRS.

1.1 Purpose

The purpose of this document is to list and describe all the requirements for developing iCOCKPIT RDP Client on iPhone and iPad. It will also explain system constraints, interface and interaction with Cockpit Server.

1.2 Intended Audience

The intended audience for this document are potential IOS Contractors participating in the bid.

1.3 Out of Scope of this document

This document provides only a general overview about COCKPIT structure, configurations and management. For more detailed information, please use the following link:
<http://jetroplatforms.com/wp-content/docs/COCKPIT%20Manual%20AD.pdf>

1.4 Definitions, acronyms, and abbreviations

We use the following typographical conventions and definitions in this document:

TERM	DEFINITION
COCKPIT	<i>Virtualized Application Delivery solution based on Microsoft terminal servers.</i>
aCOCKPIT	<i>Cockpit RDP Client for Android</i>
iCOCKPIT	<i>Cockpit RDP Client for IOS</i>
FreeRDP	<i>Free implementation of the Remote Desktop Protocol (RDP), released under the Apache license.</i>
aFreeRDP	<i>Android RDP Client based on Free RDP</i>
iFreeRDP	<i>IOS RDP Client based on Free RDP</i>
JPP	<i>Jetro Proprietary Protocol</i>

TERM	DEFINITION
COCKPIT Server	Controller or Secure Connector
Controller	COCKPIT server that manages and monitors Application Delivery on the COCKPIT site.
Secure Connector	COCKPIT server that enables remote users connecting from an unsecured network environment to establish a secure communication channel between the COCKPIT Client and the internal restricted COCKPIT4 site.
Active session	An active RDP session is a session on a server with a client connected
Idle session	An active RDP session without user input.
Disconnected session	When a session is in a disconnected state, running programs are kept active even though the user is no longer actively connected.
Reconnected session	When user logged into server with disconnected session.

1.5 Overview

The remainder of this document includes 7 chapters.

Chapter 2 provides an overview of the system functionality and system interaction with other systems. Further, the chapter also mentions the system constraints and assumption about the product.

Chapter 3 Describes the scope of the project, within which all deliverables must be produced (in the Scope) and its boundaries and restrictions (Out of Scope).

Chapter 4 provides the requirements specifications that include specific requirements, functional specifications (FS), nonfunctional specification (NFS) and design specification (DS).

Chapter 5 describes the behavior of the software

Chapter 6 describes the communication module and system infrastructures

Chapter 7 defines the deliverables of the project.

Overall description

The purpose of this section is to present a detailed description of the product's perspective giving information about the context and interface constraints. The product functions section outlines major functionality iCOCKPIT will perform. The user characteristics section explains the expectations iCOCKPIT has about the user. The constraints section contains detailed descriptions of constraints. The assumptions and dependencies section summarizes any assumptions or dependencies the application has about the hardware, software and environment.

1.6 Goals and Objectives

This project is to take the existing COCKPIT RDP Client for Android (aCOCKPIT) and convert it to COCKPIT RDP Client for IOS (iCOCKPIT). iCOCKPIT, like Android client should enable COCKPIT end users to access remotely their Windows applications directly from their iPhone and iPad devices.

1.7 Product perspective

The intent of iCOCKPIT is to utilize the functionality of COCKIT and use the system in the same way as it is implemented on windows OS and Android.

iCOCKPIT design will be based on the existing Windows COCKPIT RDP Client and aCOCKPIT that are part of Jetro client-server infrastructure for centralizing application deployment and management using Windows Terminal Services. iCOCKPIT will allow user to connect via remote desktop protocol to his remote windows desktop in the cooperate environment.

For more information about Cockpit Application Delivery Client-Server infrastructure, follow the link:

<http://jetroplatforms.com/wp-content/docs/COCKPIT%20Manual%20AD.pdf>

ICOCKPIT development will include integration with iFREE libraries, including adjusting them to meet the requirements listed in this SRS. iFree is a part of FreeRDP open source, which is a free implementation of the Remote Desktop Protocol (RDP).

For more information about FreeRDP and iFree, follow the link:

<http://www.freerdp.com/>

1.1 Product functions

There are several major functions that iCOCKPIT will perform. The most important are getting user profile, displaying data from server and manipulating the data directly via IOS device using the RDP protocol.

In the first step, the user will need to enter server connection details in order to establish the connection between iCOCKPIT and the cooperate server via JPP or SSL, depending weather this is a Controller or a Secure Connector (Gateway).

Once a connection has been established, user will be prompt to enter his credentials and email (the last one is optional). Then, iCOCKPIT will send a request to the server with user's credentials, OS, device type, and screen resolution. The Server will generate a list of settings that apply for this user and device and send the client a message with user's profile settings in JSON format. iCOCKPIT will start a remote session with start program and display graphically his published applications in their current state (non-active or reconnected).

After that all other major functions of the software can be performed. The information displayed can be easily viewed by simply navigating through the interface and selecting the desired options. The user should be able to lunch a new application by clicking an application icon and making iCOCKPIT to start an RDP session using the FreeRDP protocol. The user should be able to use several applications in a time, manipulate those using Window controls, zoom them, switch between them, minimize and close them one by one or all at once. Among others, the user should be able to send a print document to his email configured in the login screen (optional).

For future connections, iCOCKPIT should save locally the connection details and user's credentials, all except the password. The user should be able to add and save as many connections as he wants, edit the credentials related to them, and delete them one by one.

1.8 Operating Environment

The software will run on the IOS operating system version 7 and 8. All devices that support this version of the IOS operating system will be able to run the software.

1.9 Design and Implementation Constraints

- iCOCKPIT will be running on top of the iOS which is run on every iPhone/iPad. Therefore, iCOCKPIT will need to adhere to iOS specific environment capabilities and/or restrictions.
- iCOCKPIT interface will be designed based on existing IOS screens for iPhone which should be expanded and modified to meet the requirements listed in this SRS.
- iCOCKPIT must have exactly the same GUI as of RDP Client for Android as long as it adhere to iOS specific environment restrictions.
- The modified iFree libraries, screens and some other components of the software need to be open source.
- iCOCKPIT must use the language supported by the IOS development environment.
- iCOCKPIT will be developed on xCODE
- iCOCKPIT should support IOS 7 and IOS 8. The layouts' sizes should be adjusted to iPhone and iPad.

- Some iCOCKPIT functionalities will be integrated with iPhone/iPad physical Home screen button and build in menu options.
- iCOCKPIT should support fully encrypted connection over SSL and JPP protocols.
- iCOCKPIT needs to be compatible with the current iPhone operating system, starting IOS 7 and higher. iCOCKPIT needs to be compatible with all iPhone functionalities. The image sizes should be adjusted to support all iOS Devices and resolutions.
- iCOCKPIT should operate both over Wi-Fi and 3G/4G networks, as well as to support reconnect functionality.
- The data transfer between server and client should be in JSON format.

1.10 User characteristics

The user is expected to be familiar with either the iPhone or iPod touch as well as perform basic navigational and input operations. The user must belong to an organization that purchased COCKPIT licenses and fully implemented COCKPIT infrastructure on the sever side. The user must have a COCKPIT account and have organization's privileges to use iCOCKPIT. The user is also expected to be familiar with main COCKPIT functionalities.

1.11 Special usage considerations

In order to be able to use all iCOCKPIT functionalities, the only requirements besides having an IOS device will be having a COCKPIC account and internet access.

1.12 Dependency

iCOCKPIT assumes that there will be database of users' profiles hosted by a server. iCOCKPIT requires an internet connection for full functionality of the application. iCOCKPIT assumes that users have either an iPhone or iPod touch with access to the Internet from either a cellular or Wi-Fi connection to establish a connection between the application and COCKPIT server.

A major bound on the performance of iCOCKPIT is speed of the application. The speed of the application will among others depend on the network that is available in the user's location.

1.13 Work assumptions

The development process is divided into five phases that must be performed in the order specified below:

Phase 1: Compilation of Free RDP library in xCODE:
<https://github.com/FreeRDP/FreeRDP/tree/master/client/iOS>

Phase 2: Rewriting and expanding the existing communication Channel on IOS:
Implementation of Jetro proprietary Server-Client protocol.

Phase 4: Integration with existing GUI screens for iPhone and developing new once according to the design including implementation of all functionalities described in this document.

Phase 3: Making Changes in the Free RDP libraries.

Phase 5: Converting all screens for iPhone to iPad.

2 Scope of the project

This section specifies the In-Scope and Out of the Scope requirements.

2.1 In the scope

- Convert the existing Android application to IOS to meet the requirements supplied by Jetro Platforms Ltd listed in this SRS.
- Supporting any changes to this SRS that will be required by Jetro Platforms Ltd during the development period, on a Client side.
- Fixing all bugs following first production pilot.
- Successful Publishing in App Store.
- Fixing bugs following second production pilot.
- Instructing the staff and source code delivery

2.2 Out of the scope

- Fixing bugs on server side.
- Select pilot users
- Fixing mobile performance

3 Specific Requirements

This section contains all of the functional and quality requirements of the system. It gives a detailed description of the system and all its features.

3.1 General Functional requirements

Phase 1: Logging into COCKPIT account:

iCOCPIT SHOULD -

1. enable user to add many COCKPIT SSL and Direct connections
2. enable user to connect over Wi-Fi and 3G/4G networks
3. enable user to view existing COCKPIT connections' details
4. enable user to view all his existing COCKPIT connections in a list and scroll them
5. display the existing connections in alphabetical order
6. enable user to log into his COCKPIT account using any of the connections
7. enable user to define an email address to get PDF print files to his email account
8. enable user to edit his Login details and email address
9. store connection details after first successful connection to the server
10. store use's login details, except the password, after first successful login
11. enable user to delete his existing connections
12. enable user to reset his password
13. enable user to view the About screen
14. enable user to view the Help screen
15. show a progress bar during the creation of a connection and during the login.

Phase 2: Using COCKPIT remote desktop

ICOCPIT SHOULD -

1. Present Icons of all published applications based on user's profile.
2. dynamically display the most updated results of applications (refresh)
3. enable user to lunch published applications in separate screens
4. display a progress bar with cancel option when lunching first application
5. provide an advanced function keys tools
6. enable user to display the remote desktop of the first opened applications' TS
7. support seamless functionality and enable user to navigate between all running applications without closing the socket, consuming only one RDP license.
8. enable user to navigate between running applications and open new once
9. present all running applications in a side bar and allow user to scroll them

10. indicate which of the running applications is currently active
11. place an indicator next to the Icons of all currently running applications
12. enable user to close non modified running applications one by one
13. enable user to send PDF print files to his email
14. enable user to use windows controls as keyboard, touch pointer and function keys to make manipulations with RDP applications.
15. support Zoom saving in running applications regarding there state (active or non-active)
16. allow user to minimize running applications
17. enable user to close the RDP Socket
18. display a list of all running applications allowing user to decide to close the socket.
19. support the reconnect functionality
20. support network loss
21. be compatible with all other devices functionalities.

3.2 Client interfaces

Since iCOCKPIT must have almost the same GUI as of aCOCKPIT, the aCOCKIT GUI elements as well as existing iCOCKPIT GUI elements will be taken as a basis to illustrate the graphical interface required to be developed.

Most of GUI elements are presented below:

3.3 GUI general requirements

1. The following screens will be displayed in portrait orientation only:
 - Background screen (with and without progress bar)
 - Add new connection
 - Connection details screen
 - Connections screen
 - Login screen
 - Reset Password screen
 - Help screen
 - About screen
2. The following screens will be displayed in landscape and horizontal orientation:

- Home screen
- Application screen
- 3. Disabled buttons should be gray and not clickable
- 4. Enabled buttons should be orange and clickable
- 5. Tapped buttons should become dark orange and not clickable, while all other screen elements should become disabled
- 6. An active list item (connections and running application) should change its color according to the look and feel of the screen
- 7. The Keyboard:
 - Shouldn't hide the input fields/ buttons/ menus/ side bar/ on smartphones and tablets.
 - Should be raised only when tapping an editable input field
 - After being raised, the keyboard should be hidden when clicking anywhere else on the screen but other editable input field. This includes buttons, menus, text, header, etc.

For a complete GUI gap analysis between the existing screens for IOS that were already developed and the required once, please see the [Complete GAP analysis document](#).

For screens in PDF format, [please see the GUI document](#)

3.4 Specific Functional requirements

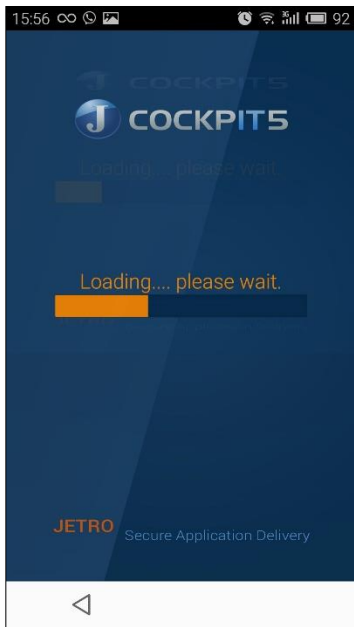
This section includes the requirements that specify all the fundamental actions of the software system

A. Background screen with/without a progress bar (portrait & view only)

This screen will be displayed in the following cases:



- Will be displayed without a progress bar after clicking the Jetro Icon for the first time after Installation (will be up less than a second before the New Connection screen is displayed).



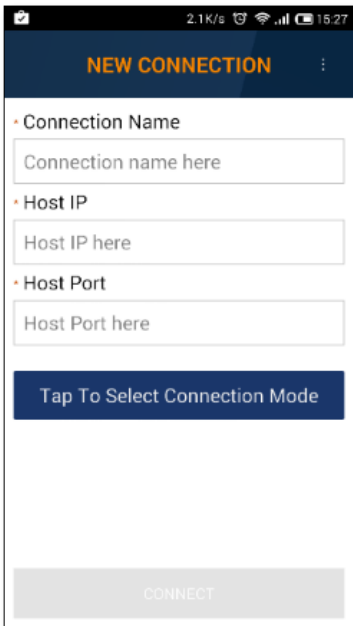
Will be displayed with a progress bar:

- During the connection, the client side will display a progress bar which freezes the client until the connection was completed or failed.
- During the login, the client side will display a progress bar which freezes the client until the connection was completed or failed.

B. New Connection screen (portrait)

The New Connection screen allows to add a new connection and it appears in the following cases:

- When the application is launched for the first time
- When there are no existing connections yet
- After user pressed on the **Add Connection** button in the Connection screen.

	Header/Footer Components: • Title: “New Connection” • Main Menu button- tapping it will display the Main menu • Back button - If there are no other existing connections, the Back button will not be displayed. - If there is at list one more existing connection, tapping the Back button will discard the details added in the fields and the Connection screen will be displayed. Connection Name field - Input text field - Title: “Connection Name” - Input field hint: Connection name here - Mandatory indicator Port field - Input numeric field of Port type - Title: Host Port - Input field hint: Host Port here - Mandatory Indicator
	Connection Mode drop down list - Hint: Tap To Select Connection Mode Tapping it will display a drop down list with 2 options: SSL and Direct. Tapping <i>outside</i> of the menu will be resulted in choosing the SSL option. Connect button Will become active when all the fields will be populated correctly.

When the New Connection screen is displayed, all fields should be empty, the hints should be displayed in the input fields, and the Connection Mode drop down list should be closed.

Pressing the Connect button will send the target server all the input data for verification.

Successful connection: Once a connection was successful, the login screen will be displayed and the new created connection will be added as a connection item into the connections list screen.

Unsuccessful connection: In case a connection was rejected by the server, a pop up message will appear with the following text: "Please check your network connection". The popup will contains the button, "ok". Clicking "ok" will close the pop up and the "New Connection" screen will be displayed with all the fields populated.

C. Connections screen (portrait)

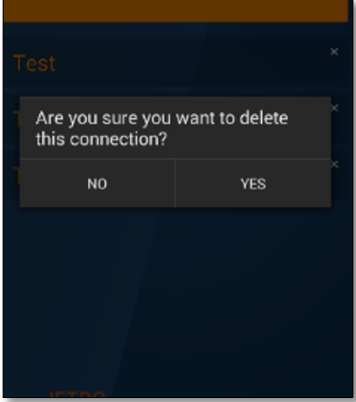
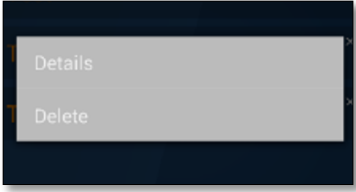
The Connections screen displays the list of all existing connections, allowing user to view their details, to delete them, to connect to the host and to add a new connection.

This screen appears in the following cases:

- Every time the application is launched, in case there is at least one existing connection item.
- After tapping Back on the New Connections, Login, Reset Password and Connection Details screens.

After Logging of RDP session and after the connection to the Controller was broken due to Network loss/other Network Issue, and after the session was disconnected locally or remotely

	Header Components: • Title: "New Connection" • Main Menu button- tapping it will display the Main menu Add connection button Tapping it will display the New Connection screen. Connection Item: Once a successful connection occurred, the connection will be added into the connection list as an item under its name. All connection items in the list should be displayed alphabetically. • Connection Item Name - the name of the connection that was created on the connection screen.
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	• Delete connection button - Clicking it will display a Delete confirmation pop up with the following text: "Are you sure you want to delete this connection?" with two options, No and Yes. - Clicking No will close the pop up, and the Connection screen will be displayed. - Clicking Yes will delete the item permanently. In case there is at least 1 additional connection in the list, the Connection screen will be displayed. In case there are no other connections, the New Connection screen will be displayed.
	Connection item menu Performing a long press on a connection item will display 2 options: - Details: Tapping it will display the Connection Details screen. - Delete: Tapping it will be resulted as if the Delete connection button was tapped.

D. Login Screen (portrait)

The Login screen prompts user to add his login details for validation. This screen will appear only after successful socket connection with COCKPIT server occurred. Note, that during the connection, the client side will display a progress bar, before the Login screen.

The Login screen will appear in the following cases:

- User filled in correctly the fields on the New Connection screen and tapped Connect.
- User tapped one of the connection items on the Connection screen.
- User tapped Login on the Connection Details screen.

The Login Screen can have two states:

- In scenario a, on first screen load all the fields will not be populated with any values, a field hint will be displayed and the login button will be disabled.
- In scenarios b and c, the fields will be populated with user's credentials and the user will have to enter only their password (configurable). All fields will be populated and editable. This because after the first successful login, all the user credentials will be saved except for the password. Saving the password will be configurable.

Header/Footer Components:

- **Title:** “Login”
- **Main Menu button**- tapping it will display the Main menu
- **Back button** - Tapping it will discard the data and the Connection screen will be displayed.

User Name field

- Input text field
- **Title:** “User Name”
- **Input field hint:** Enter user name here
- Mandatory indicator

Password field

- Input text field of Password type (***)
- **Field title:** “Password”
- **Input field hint:** Enter Password here
- Mandatory Indicator

Domain field

- Input text field
- **Field title:** “Domain”
- **Input field hint:** Enter domain here
- Mandatory Indicator

Email field (Optional)

- Input text field
- **Field title:** Email
- Input field hint: **Enter email here**

Logo Image (Optional)

After connection, a logo image will be streamed to the client, the image size will remain in each connection.

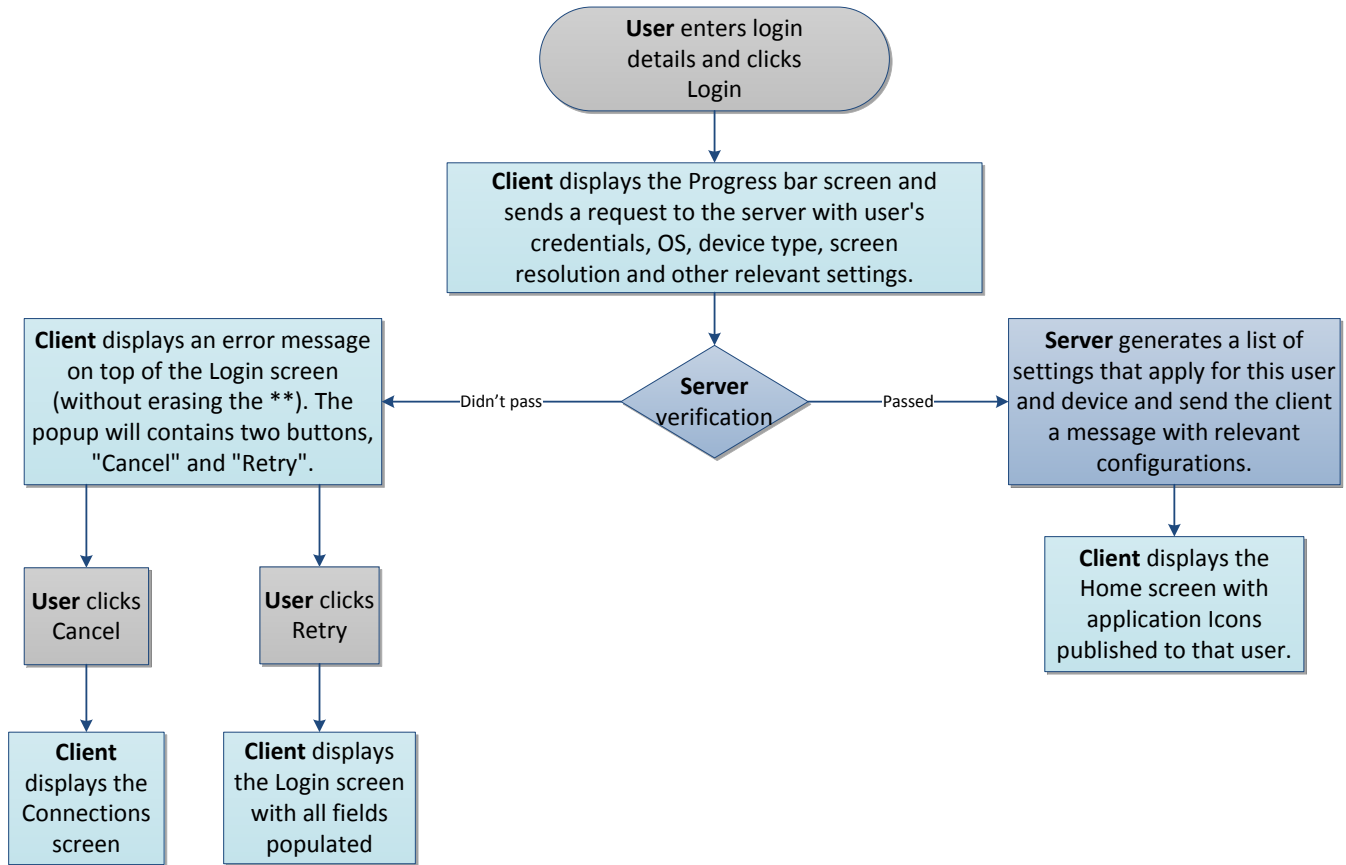
Cancel button

Should be enabled unless the Login button was tapped. Tapping the Cancel button will discard the details added in the fields and the Connection screen will be displayed.

Login button

When all Mandatory fields are populated and valid, this button becomes enabled.

User verification flow



Note, that the response for login request may contain a change password parameter which will display a Reset password Screen (E)

E. Reset Password screen (portrait)

This screen forces user to change password. The screen will appear in case Change Password is required, after Server verified user's login details. Only in case of successful password change, the user will be logged in and the Home screen will be displayed.

Header/Footer components:

- **Title** “Change Password”
- **Main Menu** - tapping it will display a menu
- **Back button** - Tapping it will discard the input data and display the Connection screen.

Old password field

- Input text field of Password type (*)
- **Field title:** “Old Password“
- **Input field hint:** Enter Old Password
- Mandatory Indicator

New Password field

- Input text field of Password type (*)
- Field title: “New Password“
- Input field hint: Enter New Password
- Mandatory Indicator

Confirm Password field

- Input text field of Password type (*)
- Field title: “Confirm New Password“
- Input field hint: Enter New Password

Reset Button

This button will be blocked until all the fields are populated with at least one character in each field.

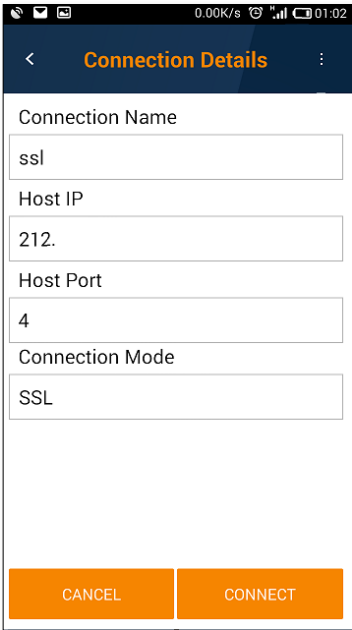
Tapping the **Reset** button will display a progress bar.

- In case of successful password change, the Home screen will be displayed and the new password will be stored on server. This will be the only password the user will be able to log into with.
- In case one of the entered parameters didn't meet the requirements, an error message will be displayed on top of the Change password screen (without erasing the *****) with OK option.

Tapping "OK" will hide the pop up and displayed the "Change Password" screen with all fields populated

F. Connection Details (portrait)

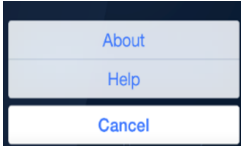
This screen displays information about existing connection (Connection Name, Host IP, Host Port and Connection Mode). The screen will appear once the user long tapped on the connection item and from the Connection menu chose the Details option. All the fields in this screen should be populated and view only.

	Header components: • Title “Connection Details” • Main Menu - tapping it will display the menu • Back button - Tapping it will display the Connections screen. Connection name field - Input text field view only - Field title: “Connection Name” Host IP field - Input text field view only - Field title: “Host IP” Host Port field - Input text field view only - Field title “Host Port” Connection Mode field - Input text field view only - Field title “Connection Mode” Cancel button Should be enabled unless the Connect button was pressed. Tapping it will display the Connections screen. Connect button Should be enabled. Tapping it will send a validation request to the server and a Progress bar will be displayed. • If the connection was rejected, a Network error message will appear with “OK” option. Tapping it will dismiss the message. • In case the connection was successful, the login screen will be displayed.
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G. Help screen (portrait)



This screen will contain a web view with HTML content (static URL) and can be opened from the main menu which can be access from the following screens: Connections, New Connection, Login, Connection Details, Reset Password, About and Help.



In case a user had pressed on this option when there is no internet connection, **an error pop up will be displayed.**

Header components:

- **Title “Help”**
- **Main Menu** - tapping it will display the menu
- **Back button** - Tapping it will display the Connections screen.

H. About screen (portrait)



This screen can be also opened from the main menu.

Header components:

- **Title “About”**
- **Main Menu** - tapping it will display the menu
- **Back button** - Tapping it will display the Connections screen.

This screen will contain **a web view with HTML content (static URL) and links.**

Only the content of the following fields should be dynamic and taken from the device:

- COCKPIT Version
- System Version
- Model

I. Home Screen (Portrait and Landscape)

The home screen will be opened in case of a successful login. This screen displays a grid of applications, published to the specific logged in user. Each application will contain an application icon and application name text. When the user had logged in, a list of published applications he may work with over RDP will be sent from the server. The application icons will be sent from the server and will be displayed properly in the grid. The application name text will be displayed beneath the application icon. Every time the home screen is displayed, an applications list request will be performed.



Application Grid

- **Background image**
- **Application icon**
 - 6*2 on smartphones.
 - 8*4 on Tablets.
 - The Icons will be reflected from the server.
- **Application name text**

Active application indicator - An orange spot will appear next to the application in case it is currently running.

Scroll bar - In case the grid will be populated with more published application than the screen can contain, a vertical scrolling will be enabled.

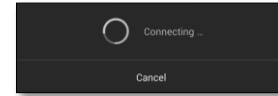
Refresh button - Pressing the Refresh will send a request for application list. During the refresh, a progress indicator will be displayed. After refresh, The updated list of currently published applications will be displayed on the Home screen, including the indicators or running applications, if exists.

Side bar - The side bar will be displayed closed by default.

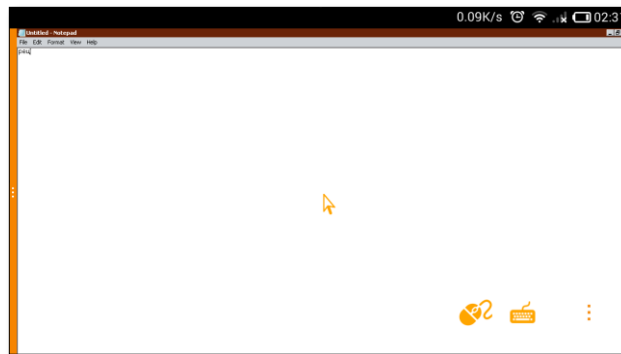
J. Application screen (Landscape and Portrait)

Tapping the Icon on the Home screen will display the Application screen.

Note, that when first time clicking on Icon, a progress bar will be displayed during the session stat. Clicking Cancel will stop the session and the Home screen will be displayed



The Application screen displays the remote desktop view. The remote desktop view allows user to view and control the remote desktop application (aka window) using ifreeRDP modified libraries.



Side bar - The side bar will be closed by default.

Keyboard and Mouse functions – As a first step, low system keyboard will be automatically shown up as well as the mouse pointer (above the keyboard) when a relevant application will be opened (based on a server message). **In the final stage of the project, Jetro custom keyboard should be developed.**

Mouse pointer – should respond to **any** screen movement.

Classic Scroll bars (vertical and horizontal) – will be automatically displayed when user will perform zoom in, and in a webpage.

Tool Bar – should be floating. By default, the tool bar should appear on top right corner of the screen and include 3-5 icons. Tapping them will display/dismiss the relevant option:

- **Keyboard** – tapping it will display a low system keyboard (or dismiss it in case it was up). Tapping it again will dismiss the keyboard. Opening the side bar should hide the keyboard in case it was up.
- **Mouse button** – tapping it will display a mouse pointer (or dismiss it in case it was up) as well as an additional button in the toolbar:
- **Move button** – tapping it should enable user to drag the Tool bar.
- **Advance Scroll button** – tapping it will display an icon with 4 errors (options). Should be draggable.

- **Tools Icon** - should include the following functions (Icons): **Should be developed in the final stage of the project (Optional).**

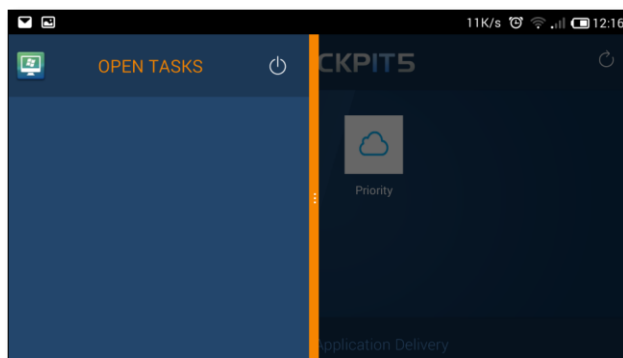
Function keys, Zoom in – in one click (full screen), Page Up & Page Down (Active only in Web pages and after Zoom out Back & Forward (Active/Available in Web pages), Forward (Active/Available in Web Pages), Zoom squares – dividing the screen to squares. Tapping any of the squares will show it in a full screen, Others, like "Copy", "Paste", "Print", "Save".

Zoom - By default application should be opened in a full screen. Zoom In and Out should be saved per application, and be implemented in all screen states, regardless which function is currently active (mouse, keyboard, ect).

K. Side bar

The Side bar allows users to Log off the session, to display the Remote Desktop view of the first launched application, to view all running applications, to navigate between them and the Home screen, as well as to close them one by one.

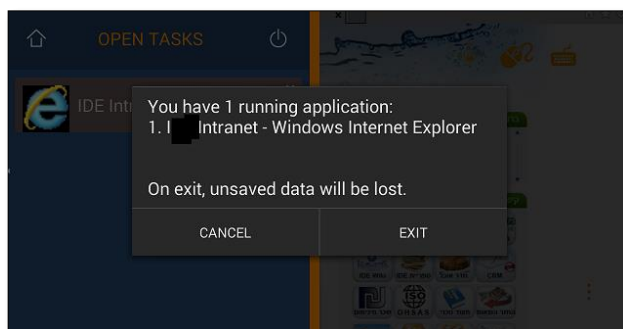
The Side bar can be accessed from the Home and from the Application screens.

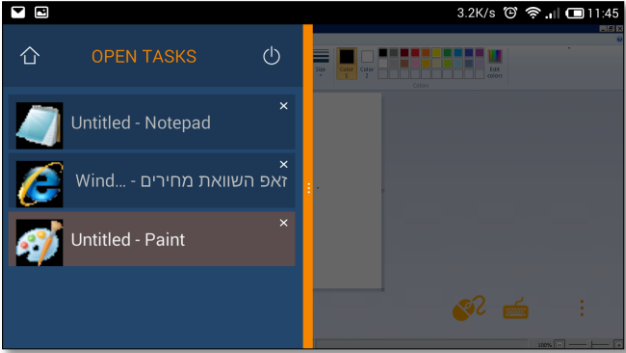


RDP button - will be displayed on side bar only during first application is launched, and tapping it will display the Remote Desktop.

Log off button – will be always displayed on side bar.

- In case there are no running applications, tapping it will close the socket and the home screen will be displayed.
- In case there is at least one running application a message with all running applications will be displayed, including two options: Cancel and Exit.
 - **Cancel** – tapping it will close the message
 - **Exit** – tapping it will close the socket by performing a force log off. Meaning all applications will be closed, no changes will be saved and the Home screen will be displayed.





Application list – each running applications appear as a list item in the side bar applications list. Tapping on a list item will make that application window to be active.

Application item – includes an Icon, application name, a document name/URL, and a Close button.

Home button - Will be displayed only if the Side bar was opened from the Application screen. Pressing it will display the Home screen.

Other Scenarios:

Minimizing	
User minimizes application X window and has other running application Y	The app X is minimized, and other running app Y is displayed in the Application window. Application Y is also appears as "Active" in the task bar. The minimized app X is still running therefore the application appears on the task bar as well as indicated on the Home screen as "running".
User minimizes last running application	The Home screen is displayed.
Closing Applications one by one	
User closes application X window and has other running application Y (no unsaved changes)	The app X is closed, and other running app Y is displayed in the Application window. Application Y is also appears as "Active" in the task bar. The closed app X is removed from the task bar as well as its "running" indicator is disappeared from the Home screen.
User closes last running application (no unsaved changes)	The app is closed and the Home screen is displayed. The closed app is disappeared from the task bar as well as its "running" indicator is disappeared from the Home screen.

Users tries to close a running application with unsaved changes	The local message “Do you want to save changes” will be displayed, and the user won’t be able to close this application without responding to this message. <i>Note that the process can be always terminated by closing the socket with the help of the Log off function.</i>
Changing Application's title	
User changes the title of the application (Saves/open IE page)	The new title should be displayed on the side bar
Navigating between running applications using the Side bar	
User taps an active application	The application is displayed in the Application window
User taps application that is not active	The application is displayed in the Application window
Sending print PRF files to email defined in the Login screen	
User taps Print document	The user will receive a PDF file to the email that he defined in the Login screen.

3.5 Non-Functional requirements

3.5.1 Supporting Network Loss

During all the time when the socket stays open, the Server will send a keep alive messages every 15 sec. In case the Client won’t get 3 keep alive messages, an error message should be displayed with the following text: “The network connection was lost. Please try again later” with option “Exit”. Clicking this option will display the Connection screen. Note, that in case of a connection failure, the session will get a Disconnected state.

3.5.2 Compatibility with all IOS Devices functions

Using IOS Home button shouldn’t affect iCOCKPIT behavior. Meaning, moving iCOCKPIT to the/from the background processes shouldn’t have any impact on any of iCOCKPIT functions & states. This also applies to screen saver, custom keyboards, ringing, recording and all other device functions and settings.

3.5.3 Supporting Reconnect function

In case the socket was closed unexpectedly (not via Log off function) due to network loss, battery discharge, killing client from running background processes, timeout, session disconnected, when the user will re-login the reconnected session should be automatically displayed.

3.5.4 Security

The username and the password should be encrypted. The connection details and the user credentials - all will be saved by default, except from the password. The password will be saved only in case the server will send a relevant message.

The data transfer between Client and Server should be enabled both, over SSL and JPP.

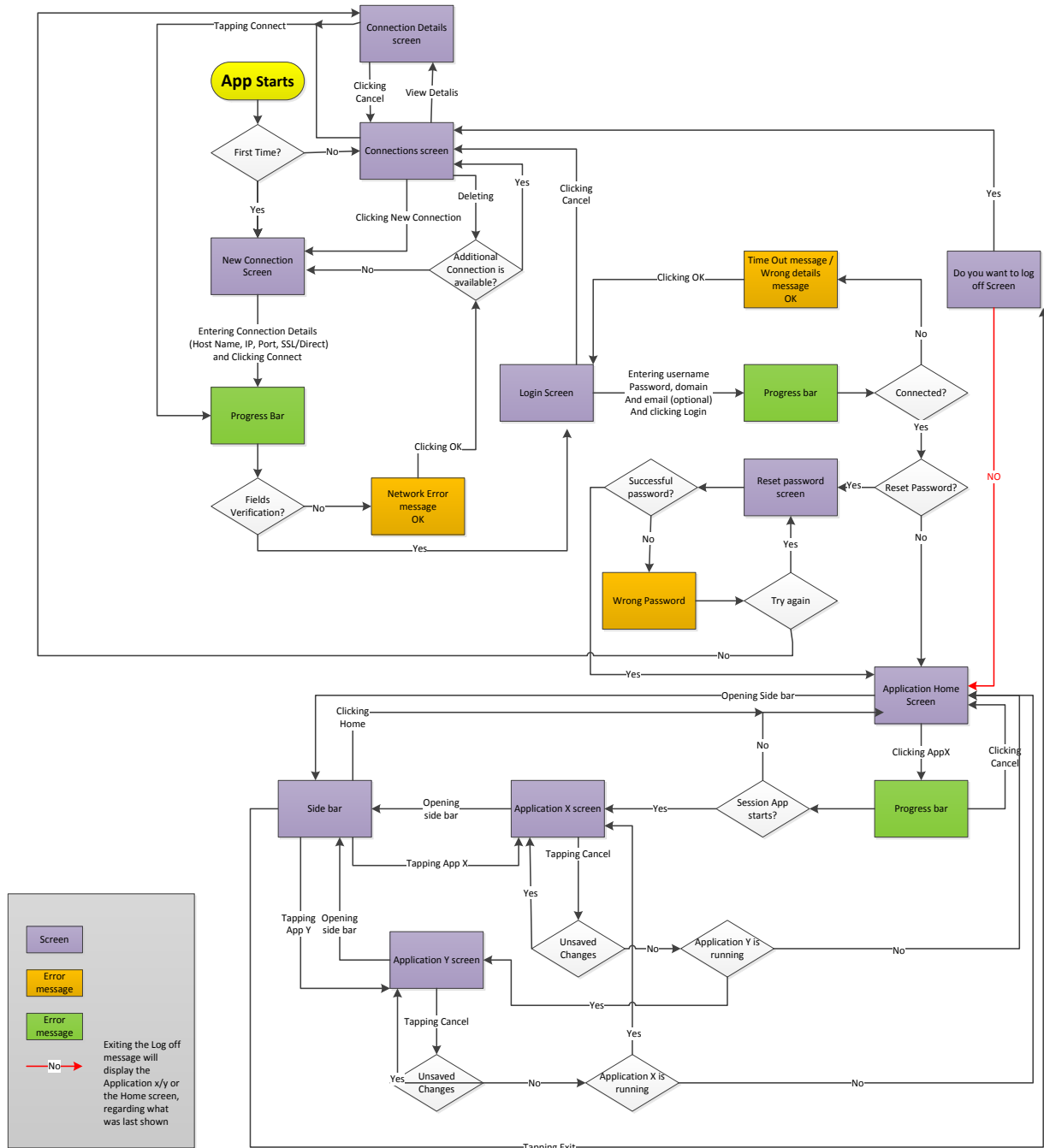
3.5.5 Performance

iCOCKPIT performance, capacity and reliability should be equal or better than iFree.

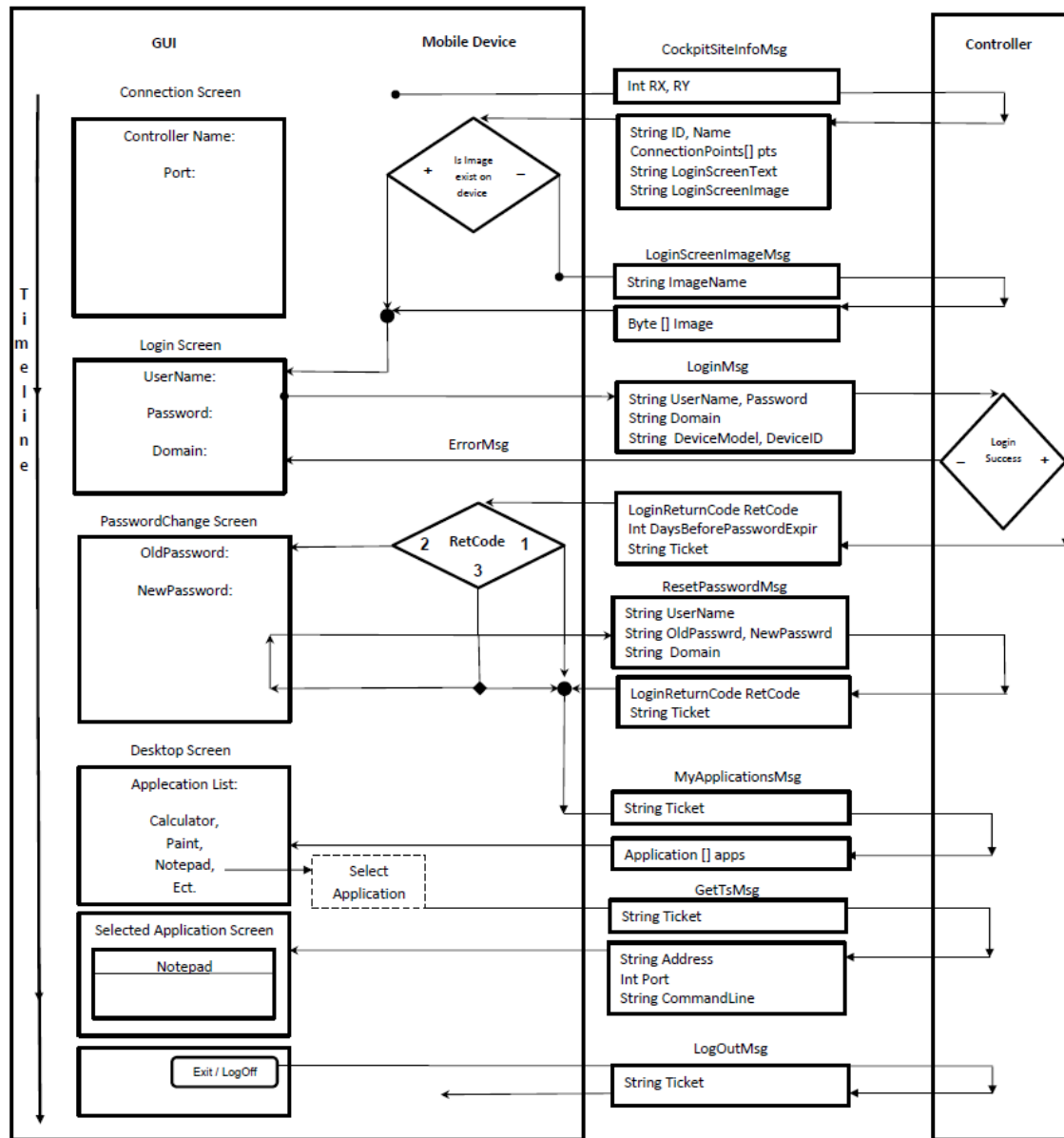
3.5.6 Coding style

The Code should be written in the most clean, easy to read and maintain manner, according to Apple's coding requirements.

4 Behavioral Model



5 Communication



6 Deliverables

The acceptance criteria of the project are stated below:

- Completed on time, within the budget and with acceptable agreed upon level of quality. (As per requirements - all works as expected)
- Documentation completed
- Training provided
- Sources provided
- iCOCKPIT is approved by Apple and can be downloaded from AppStore.

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